



**SYDNEY BOYS HIGH
SCHOOL
MOORE PARK, SURRY HILLS**

**2003
TRIAL HIGHER SCHOOL
CERTIFICATE**

Mathematics Extension 2

General Instructions

- Reading Time – 5 Minutes
- Working time – 3 hours
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators may be used.
- All necessary working should be shown in every question.

Total Marks – 120

- Attempt all questions.
- All questions are of equal value.
- Each section is to be answered in a separate bundle, labeled Section A (Questions 1, 2, 3), Section B (Questions 4, 5, 6) and Section C (Questions 7 and 8).

Examiner: *C.Kourtesis*

Note: This is an assessment task only and does not necessarily reflect the content or format of the Higher School Certificate.

Section A
Start a new answer sheet

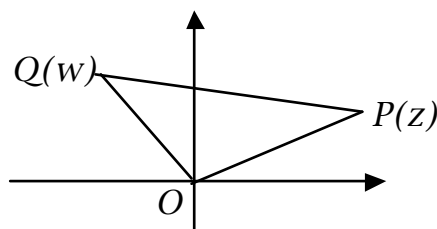
Question 1. (Start a new answer sheet.) (15 marks)

	Marks
(a) Find $\int \frac{dx}{\sqrt{4-9x^2}}$.	2
(b) Find $\int \frac{4}{(x-1)(2-x)} dx$	3
(c) Use integration by parts to find $\int te^{\frac{t}{4}} dt$	3
(d) Use the substitution $u = 2 + \cos \theta$ to show that $\int_0^{\frac{\pi}{2}} \frac{\sin 2\theta}{2 + \cos \theta} d\theta = 2 + 4 \log_e \left(\frac{2}{3} \right)$	4
(e) Evaluate $\int_0^{2\pi} \sin x dx$	2
(f) Determine whether the following statement is True or False, and give a brief reason for your answer. $\int_{-1}^4 \frac{dx}{x^3} = \frac{15}{32}$	1

Question 2. (15 marks)

- | | Marks |
|---|--------------|
| (a) (i) Express $w = -1 - i$ in modulus-argument form. | 2 |
| (ii) Hence express w^{12} in the form $x + iy$ where x and y are real numbers. | 2 |
| (b) Find the equation, in Cartesian form, of the locus of the point z if
$ z - i = z + 3 .$ | 2 |
| (c) Sketch the region in the Argand diagram that satisfies the inequality
$\operatorname{Re}\left(\frac{1}{z}\right) \leq \frac{1}{2}.$ | 3 |
| (d) (i) On the Argand diagram draw a neat sketch of the locus specified by
$\arg(z + 1) = \frac{\pi}{3}.$ | 1 |
| (ii) Hence find z so that $ z $ is a minimum. | 2 |
| (e) Points P and Q represent the complex numbers z and w respectively in the Argand Diagram.

If $\triangle OPQ$ (where O is the origin) is equilateral | |
| (i) Explain why $wz = z^2 \operatorname{cis} \frac{\pi}{3}$. | 1 |
| (ii) Prove that $z^2 + w^2 = zw$. | 2 |



Question 3. (15 marks)

	Marks
(a) Sketch the following curves on separate diagrams, for $-\frac{3\pi}{2} \leq x \leq \frac{3\pi}{2}$. [Note: There is no need to use calculus.]	
(i) $y = \tan x$	1
(ii) $y = \tan x $	1
(iii) $y = \tan x $	1
(iv) $y = \tan^2 x$	2
(b) Consider the function $f(x) = \frac{x}{\ln x}$, $x > 0$	
(i) Determine the domain and write down the equations of any asymptotes.	2
(ii) Show that there is a minimum turning point at (e, e) .	3
(iii) Show that there is a point of inflexion at $x = e^2$.	3
(iv) Sketch the graph of $y = f(x)$.	2

Section B
Start a new booklet.

Question 4 (15 marks)

	Marks
(a) (i) By solving the equation $z^3 = 1$ find the three cube roots of 1.	2
(ii) Let w be a cube root of 1 where w is not real. Show that $1 + w + w^2 = 0$.	1
(iii) Find the quadratic equation, with integer coefficients, that has roots $4 + w$ and $4 + w^2$.	3
(b) A monic cubic polynomial, when divided by $x^2 + 4$ leaves a remainder of $x + 8$ and when divided by x leaves a remainder of -4 . Find the polynomial in expanded form.	3
(c) Consider the polynomial $P(z) = z^3 + az^2 + bz + c$ where a , b and c are all real. If $P(\theta i) = 0$ where θ is real and non-zero:	
(i) Explain why $P(-\theta i) = 0$	1
(ii) Show that $P(z)$ has one real zero.	1
(iii) Hence show that $c = ab$, where $b > 0$.	4

Question 5 (15 marks)**Marks**

- (a) A particle of mass m falls vertically from rest at a height of H metres above the Earth's surface, against a resistance mkv when its speed is v m/s. (k is a positive constant).
Let x m be the distance the particle has fallen, and v m/s its speed at x . Let g m/s² be the acceleration due to gravity.

- (i) Show that the equation of motion is given by

1

$$v \frac{dv}{dx} = g - kv$$

- (ii) If the particle reaches the surface of the Earth with speed V_0 , show that

4

$$\ln\left(1 - \frac{kV_0}{g}\right) + \frac{kV_0}{g} + \frac{k^2H}{g} = 0.$$

- (iii) Show that the time T taken to reach the Earth's surface is given by

3

$$T = \frac{1}{k} \ln\left(\frac{g}{g - kV_0}\right).$$

- (iv) Show that $V_0 = Tg - kH$.

2

- (v) Hence prove that $T < \frac{1}{k} + \frac{kH}{g}$.

1

- (b) The letters A, B, C, D, E, F, I, O are arranged in a circle. In how many ways can this be done if at least two of the vowels are together?

2

- (c) A man has five friends. In how many ways can he invite one or more of them to dinner?

2

Question 6 (15 marks)

		Marks
(a)	(i) Expand $(\cos\theta + i\sin\theta)^3$ and hence express $\cos 3\theta$ and $\sin 3\theta$ in terms of $\cos\theta$ and $\sin\theta$ respectively.	2
	(ii) Show that $\cot 3\theta = \frac{t^3 - 3t}{3t^2 - 1}$ where $t = \cot\theta$.	2
	(iii) Solve $\cot 3\theta = 1$ for $0 \leq \theta \leq 2\pi$.	2
	(iv) Hence show that $\cot \frac{\pi}{12} \cdot \cot \frac{5\pi}{12} \cdot \cot \frac{9\pi}{12} = -1$.	2
	(v) Write down a cubic equation with roots $\tan \frac{\pi}{12}$, $\tan \frac{5\pi}{12}$, $\tan \frac{9\pi}{12}$.	1
	[Express your answer as a polynomial equation with integer coefficients.]	
(b)	(i) Draw a sketch showing that if $f(x)$ and $g(x)$ are continuous functions and $f(x) > g(x) > 0$ for $a \leq x \leq b$ then	2
	$\int_a^b f(x) dx > \int_a^b g(x) dx.$	
	(ii) Show that $y = \tan x$ is an increasing function for $\frac{\pi}{4} \leq x \leq \frac{\pi}{3}$.	1
	(iii) Prove that $\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\tan x}{x} dx > \log_e \left(\frac{4}{3} \right)$.	3

Section C
Start a new booklet

Question 7 (15 marks)

Marks

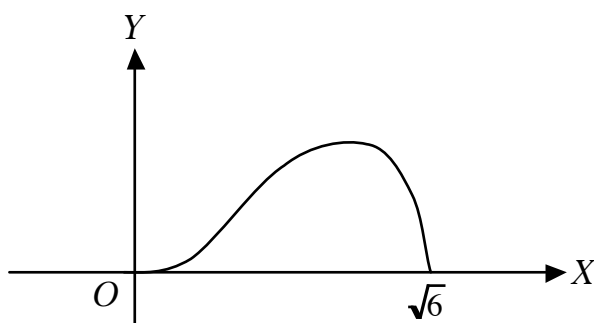
- (a)
- (i) If $I_n = \int_1^e x(\ln x)^n dx$ (where n is a non-negative integer)
- show that $I_n = \frac{e^2}{2} - \frac{n}{2} I_{n-1}$ (where $n \geq 1$).
- (ii) Hence evaluate I_3 .

3

2

(b)

4

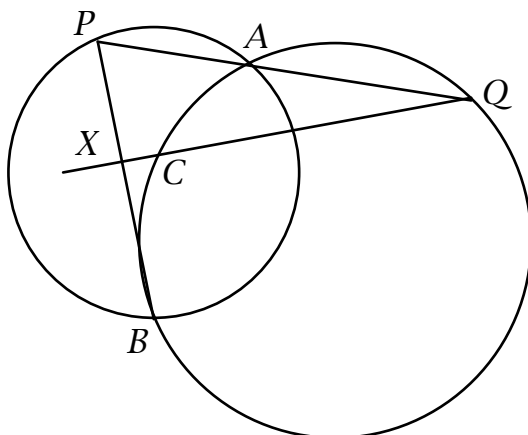


The diagram shows the graph of $y = x^2(6 - x^2)$ for $0 \leq x \leq \sqrt{6}$. The area bounded by this curve and the x -axis is rotated through one revolution about the y -axis.

Use the method of cylindrical shells to find the volume of the solid that is generated.

Question continued

(c)



The two circles intersect at A and B . The larger circle passes through the centre C of the smaller circle. P and Q are points on the circles such that PQ passes through A . QC is produced to meet PB at X .

Let $\angle QAB = \theta$.

(i) Make a neat copy of the diagram on your answer sheet.

(ii) Show that $\angle BCX = 180^\circ - \theta$.

2

(iii) Prove that $\angle PXC = 90^\circ$.

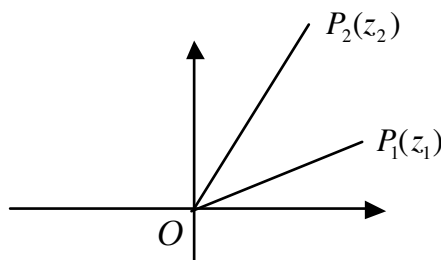
4

Question 8 (15 marks)**Marks**
4

- (a) Two of the roots of $x^3 + ax^2 + bx + c = 0$ are α and β .

Prove that $\alpha\beta$ is a root of $x^3 - bx^2 + acx - c^2 = 0$.

- (b) The points P_1 and P_2 represent the complex numbers z_1 and z_2 on the Argand diagram.



- (i) Prove that $|z_1 - z_2| \geq |z_1| - |z_2|$ 2

- (ii) If $\left|z - \frac{4}{z}\right| = 2$ prove that the maximum value of $|z|$ is $\sqrt{5} + 1$. 3

- (c) (i) Prove that if the polynomial equation $P(x) = 0$ has a root of multiplicity n , then the derived polynomial equation $P'(x) = 0$ has the same root with multiplicity $n - 1$. 2

- (ii) If the equation $x^3 + 3px^2 + 3qx + r = 0$ has a repeated root, show that this root is $\frac{r - pq}{2(p^2 - q)}$, where $p^2 \neq q$. 4

This is the end of the paper.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax \, dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax \, dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax \, dx = \frac{1}{a} \tan ax,$$

$$\int \sec ax \tan ax \, dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

$$\text{NOTE: } \ln x = \log_e x, \quad x > 0$$



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Mathematics Extension 2

Sample Solutions

Question 1.

$$(a) \int \frac{dx}{\sqrt{4-9x^2}} = \frac{1}{3} \int \frac{dx}{\sqrt{\frac{4}{9}-x^2}} \quad (2)$$

$$= \frac{1}{3} \sin^{-1} \frac{3x}{2} + C$$

$$(b) \text{ Let } \frac{4}{(x-1)(2-x)} = \frac{A}{x-1} + \frac{B}{2-x}$$

$$\therefore 4 = A(2-x) + B(x-1)$$

$$\text{If } x=2: 4 = B$$

$$\text{If } x=1: 4 = A.$$

$$\therefore \int \frac{4}{(x-1)(2-x)} dx = 4 \left(\int \frac{dx}{x-1} - \int \frac{dx}{x-2} \right)$$

$$= 4 (\ln(x-1) - \ln(x-2))$$

$$= 4 \ln \left(\frac{x-1}{x-2} \right) + C \quad (3)$$

$$(c) \int t e^{\frac{t}{4}} dt \quad u=t \quad v=4e^{\frac{t}{4}}$$

$$u'=1 \quad v'=e^{\frac{t}{4}}$$

$$= 4te^{\frac{t}{4}} - \int 4e^{\frac{t}{4}} dt \quad (3)$$

$$= 4te^{\frac{t}{4}} - 16e^{\frac{t}{4}} + C$$

$$(d) \int_0^{\pi/2} \frac{5 \sin 2\theta}{2 + \cos \theta} d\theta$$

Let $u = 2 + \cos \theta$

$$= \int_0^{\pi/2} \frac{2 \sin \theta \cos \theta}{2 + \cos \theta} d\theta \quad du = -\sin \theta d\theta$$

If $\theta=0 \quad u=3$
If $\theta=\pi/2 \quad u=2$

$$= 2 \int_3^2 \frac{u-2}{u} (-du)$$

$$= 2 \int_2^3 \left(1 - \frac{2}{u} \right) du \quad (4)$$

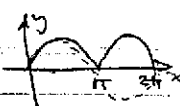
$$= 2 [u - 2 \ln u]_2^3$$

$$= 2 \{ [3 - 2 \ln 3] - [2 - 2 \ln 2] \}$$

$$= 2 \{ 1 + 2 \ln 2 - 2 \ln 3 \}$$

$$= 2 \{ 1 + 2 \ln \frac{2}{3} \}$$

$$= 2 + 4 \ln \frac{2}{3}$$

$$(e) \int_0^{2\pi} |\sin x| dx$$


$$= 2 \int_0^{\pi} \sin x dx$$

$$= 2 [-\cos x]_0^{\pi}$$

$$= 2 [-1] - [-1]$$

$$= 2 \times 2$$

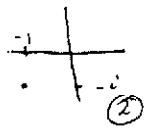
$$= 4 \quad (2)$$

$$(f) \int_{-1}^4 \frac{dy}{x^3}$$


Integral is not defined due to discontinuity at $x=0$. (1)

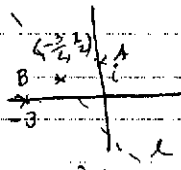
Question 2:

(a) (i) $w = -1 - i$
 $= \sqrt{2} \operatorname{cis}(-\frac{3\pi}{4})$



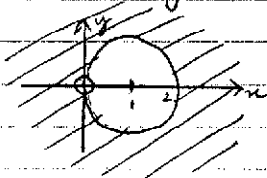
(ii) $w^{12} = (\sqrt{2} \operatorname{cis}(-\frac{3\pi}{4}))^{12}$
 $= 2^6 \operatorname{cis}(-\frac{9\pi}{2})$
 $= 64 \operatorname{cis}(-9\pi)$
 $= 64 \operatorname{cis}\pi$
 $= -64$

(b) $|z - i| = |z + 3|$

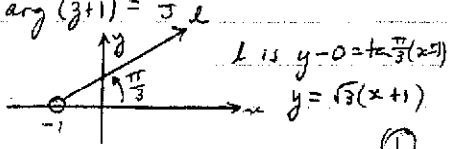


$m_{AB} = \frac{1}{3}$
 $m_L = -3$
 $\therefore$ locus is $y - \frac{1}{2} = -3(x + \frac{3}{2})$
 $y - \frac{1}{2} = -3x - \frac{9}{2}$
 $y = -3x - 4$

(c) $\operatorname{Re}(\frac{1}{z}) \leq \frac{1}{2} \quad (z \neq 0)$
 $\operatorname{Re}(\frac{x-iy}{x^2+y^2}) \leq \frac{1}{2}$
 $\operatorname{Re}(\frac{x-iy}{x^2+y^2}) \leq \frac{1}{2}$
 $\therefore \frac{x}{x^2+y^2} \leq \frac{1}{2}$
 $\therefore 2x \leq x^2+y^2$
 $\therefore x^2-2x+1+y^2 \geq 1$
 $\therefore (x-1)^2+y^2 \geq 1 \quad (z \neq 0)$



(d) (i) $\arg(z+1) = \frac{\pi}{3}$



L is $y - 0 = \tan(\frac{\pi}{3})(x+1)$
 $y = \sqrt{3}(x+1)$

(ii) $|z|$ is a minimum at A where
 $OA \perp L$

A is $(a, \sqrt{3}(a+1))$

$m_{OA} = \frac{\sqrt{3}(a+1)}{a} = -\frac{1}{\sqrt{3}}$

$\therefore 3(a+1) = -a$

$\therefore 4a+3 = 0$

$\therefore a = -\frac{3}{4}$

$\therefore A$ is $(-\frac{3}{4}, \frac{\sqrt{3}}{4})$

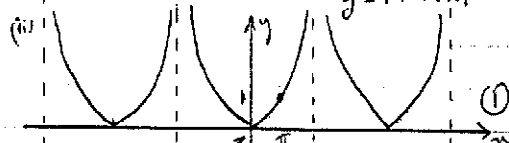
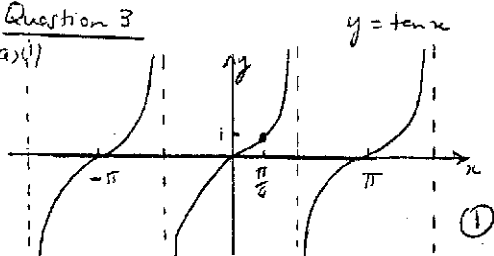
$\therefore z = -\frac{3}{4} + i\frac{\sqrt{3}}{4}$ is the required solution

(a) (i) $w = z \operatorname{cis} \frac{\pi}{3}$
 $\text{or } |w| = |z| \text{ and } \angle QOP = \frac{\pi}{3} \text{ (equilateral)}$
 $\therefore wz = (z \operatorname{cis} \frac{\pi}{3}) \cdot z$
 $= z^2 \operatorname{cis} \frac{\pi}{3}$

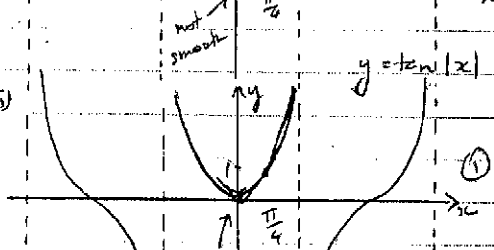
(ii) $z^2 + w^2 = z^2 + z^2 \operatorname{cis} \frac{2\pi}{3}$
 $= z^2(1 + \operatorname{cis} \frac{2\pi}{3})$
 $= z^2(1 + \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3})$
 $= z^2(1 - \frac{1}{2} + i \frac{\sqrt{3}}{2})$
 $= z^2(\frac{1}{2} + i \frac{\sqrt{3}}{2})$
 $= z^2 \operatorname{cis} \frac{\pi}{3}$
 $= wz$

Question 3

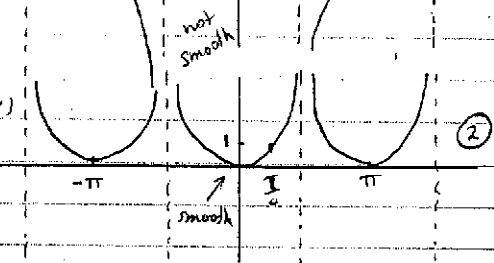
(a)(i)



(ii)



(iv)



(b) $f(x) = \frac{x}{\ln x}$, $x > 0$

(i) Domain $0 < x < 1, x > 1$

Asymptote $x = 1$

(ii) $f'(x) = \frac{\ln x \cdot 1 - x \cdot \frac{1}{x}}{(\ln x)^2}$

$= \frac{\ln x - 1}{(\ln x)^2}$

$f''(x) = \frac{(\ln x)^2 \cdot \frac{1}{x} - (\ln x - 1) \cdot 2 \ln x \cdot \frac{1}{x}}{(\ln x)^4}$

$= \frac{\ln x - 2 \ln x + 2}{x (\ln x)^3}$

$= \frac{2 - \ln x}{x (\ln x)^3}$

For turning point: $\ln x - 1 = 0$

$\therefore \ln x = 1$

$\therefore x = e$

$\therefore y = \frac{e}{\ln e} = e$

$y'' = \frac{2 - \ln e}{e (\ln e)^3}$

$= \frac{2 - 1}{e \cdot 1}$

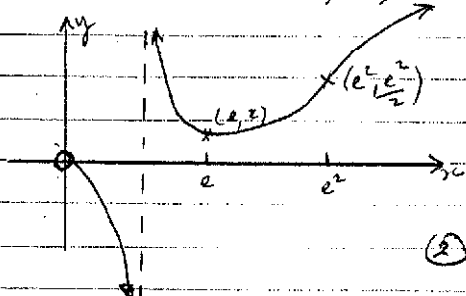
$= \frac{1}{e} > 0$

$\therefore$ Min turning point at (e, e)

(iii) x	$e^2 - \epsilon$	e^2	$e^2 + \epsilon$
y''	$\frac{2 - \ln(e^2 - \epsilon)}{(e^2 - \epsilon) \ln(e^2 - \epsilon)^3}$ $= \frac{+ve}{+ve \cdot +ve}$ $= +ve$	$\frac{2 - \ln e^2}{e^2 (\ln e^2)^3}$ $= 0$	$\frac{2 - \ln(e^2 + \epsilon)}{(e^2 + \epsilon) \ln(e^2 + \epsilon)^3}$ $= \frac{-ve}{+ve \cdot +ve}$ $= -ve$
Concavity	up	-	down

$\therefore$ Change of concavity at $x = e^2$

$\therefore$ Pt of inflexion at $(e^2, \frac{e^2}{2})$



$$\begin{aligned}
 [2] \quad 4. \quad (a) \quad (i) \quad & z^3 - 1 = 0, \\
 & (z-1)(z^2 + z + 1) = 0, \\
 & \therefore z = 1 \text{ or } \frac{-1 \pm \sqrt{1-4}}{2}, \\
 & = 1 \text{ or } \frac{-1 \pm \sqrt{3}i}{2}.
 \end{aligned}$$

$$[1] \quad (ii) \quad (\omega-1)(\omega^2 + \omega + 1) = 0 \text{ from (i).} \\
 \text{Now } \omega \neq 1, \therefore \omega^2 + \omega + 1 = 0.$$

$$\begin{aligned}
 [3] \quad (iii) \quad & \alpha + \beta = 4 + \omega + 4 + \omega^2, \\
 & = 7 + 1 + \omega + \omega^2, \\
 & = 7. \\
 & \alpha\beta = 16 + 4\omega + 4\omega^2 + \omega^3, \\
 & = 12 + 4(1 + \omega + \omega^2) + 1, \\
 & = 13. \\
 \therefore & z^2 - 7z + 13 = 0.
 \end{aligned}$$

$$\begin{aligned}
 [3] \quad (b) \quad & P(x) = x^3 + ax^2 + bx + c. \\
 & P(0) = c = 4. \\
 & P(x) = (x^2 + 4)(x + \alpha) + x + 8. \\
 & P(0) = 4\alpha + 8 = -4, \\
 & \alpha = -3. \\
 & (x^2 + 4)(x - 3) + 8 = x^3 - 3x^2 + 4x - 12 + x + 8. \\
 \therefore & P(x) = x^3 - 3x^2 + 5x - 4.
 \end{aligned}$$

$$[1] \quad (c) \quad (i) \quad \text{As the polynomial has real coefficients, if } (x - i\theta) \text{ is a factor} \\
 \text{then } (x + i\theta) \text{ is also a factor (conjugate root theorem).} \\
 \text{i.e., } P(-i\theta) = 0.$$

$$\begin{aligned}
 [1] \quad (ii) \quad & (z^2 + \theta^2) \text{ is a factor of } P(x). \text{ Let } (z - \alpha) \text{ be the last factor.} \\
 & \text{Sum of roots is } \theta i - \theta i + \alpha = -a. \\
 & \text{i.e., } \alpha = -a \text{ which is real,} \\
 \therefore & \alpha \text{ is real and there is one real root.}
 \end{aligned}$$

$$\begin{aligned}
 [4] \quad (iii) \quad & \text{Taking roots two at a time,} \\
 & b = \theta^2 + \theta ia - \theta ia, \\
 & = \theta^2. \\
 \therefore & b > 0 \text{ as } \theta \in \mathbb{R}. \\
 \text{Product of roots, } & -c = -\theta^2 i^2 (-a), \\
 & c = \theta^2 a, \\
 & = ab.
 \end{aligned}$$

$$\begin{aligned}
 [1] \quad 5. \quad (a) \quad (i) \quad & \begin{array}{c} \uparrow mkv \\ 0 \text{---} \bullet \text{---} \downarrow mg \\ | \\ H \text{---} \cdots \cdots \cdots \\ \downarrow x \end{array} \quad \begin{aligned} \sum F_x &= ma = mg - mkv, \\ a &= g - kv. \\ \text{i.e. } v \frac{dv}{dx} &= g - kv. \end{aligned} \end{aligned}$$

$$\begin{aligned}
\boxed{4} \quad (ii) \quad \int dx &= \int \frac{v \, dv}{g - kv}, \\
&= -\frac{1}{k} \int \frac{g - kv}{g - kv} dv + \frac{-g}{k^2} \int \frac{-k \, dv}{g - kv}, \\
x &= -\frac{v}{k} - \frac{g}{k^2} \ln(g - kv) + c. \\
\text{When } x &= 0, v = 0, \therefore c = \frac{g}{k^2} \ln g. \\
x &= \frac{g}{k^2} \ln \left(\frac{g}{g - kv} \right) - \frac{v}{k}. \\
\text{When } x &= H, v = V_0, \\
H &= \frac{g}{k^2} \ln \left(\frac{g}{g - kV_0} \right) - \frac{V_0}{k}. \\
\text{Rearranging, } \ln \left(\frac{g - kV_0}{g} \right) + \frac{kV_0}{g} + \frac{k^2 H}{g} &= 0, \\
\text{i.e., } \ln \left(1 - \frac{kV_0}{g} \right) + \frac{kV_0}{g} + \frac{k^2 H}{g} &= 0.
\end{aligned}$$

$$\begin{aligned}
\boxed{3} \quad (iii) \quad \frac{dv}{dt} &= g - kv, \\
\int dt &= \frac{-1}{k} \int \frac{-k \, dv}{g - kv}, \\
t &= -\frac{1}{k} \ln(g - kv) + c. \\
\text{When } t &= 0, v = 0, \therefore c = \frac{1}{k} \ln g, \\
\text{So } t &= \frac{1}{k} \ln \left(\frac{g}{g - kv} \right). \\
\text{When } t &= T, v = V_0, \\
\therefore T &= \frac{1}{k} \ln \left(\frac{g}{g - kV_0} \right).
\end{aligned}$$

$$\begin{aligned}
\boxed{2} \quad (iv) \quad \ln \left(1 - \frac{kV_0}{g} \right) &= -kT \text{ from (iii).} \\
\text{Substitute in (ii),} \\
-kT + \frac{kV_0}{g} + \frac{k^2 H}{g} &= 0, \\
\frac{kV_0}{g} &= kT - \frac{k^2 H}{g}, \\
V_0 &= gT - kH.
\end{aligned}$$

$$\begin{aligned}
\boxed{1} \quad (v) \quad \text{Terminal velocity occurs when } \dot{x} &= 0, \\
\text{i.e. } V_T &= \frac{g}{k}. \\
\text{Now } V_0 &< V_T, \\
\therefore V_0 &< \frac{g}{k} \\
T &= \frac{V_0}{g} + \frac{kH}{g} \text{ from (iv),} \\
T &< \frac{g}{k} \times \frac{1}{g} + \frac{kH}{g}, \\
T &< \frac{1}{k} + \frac{kH}{g}.
\end{aligned}$$

- 2 (b) At least two together $\Rightarrow$ not all separate.
 Total number of arrangements in a circle = $7!$
 Number of arrangements where separated = $3!4!$
 $\therefore$ Ways with at least two together = $7! - 3!4!$
 $= 4896$.

2 (c) Number of ways = $\binom{5}{1} + \binom{5}{2} + \binom{5}{3} + \binom{5}{4} + \binom{5}{5}$,
 $= 2^5 - 1$,
 $= 31$.

2 6. (a) (i) $\cos 3\theta = (\cos \theta)^3$, by De Moivre's Theorem.
i.e., $\cos 3\theta + i \sin 3\theta = \cos^3 \theta + 3i \sin \theta \cos^2 \theta + 3i^2 \sin^2 \theta \cos \theta + i^3 \sin^3 \theta$.
 Equating real and imaginary parts,
 $\cos 3\theta = \cos^3 \theta - 3 \sin^2 \theta \cos \theta$,
 $= \cos^3 \theta - 3(1 - \cos^2 \theta) \cos \theta$,
 $= \cos^3 \theta - 3 \cos \theta + 3 \cos^3 \theta$,
 $= 4 \cos^3 \theta - 3 \cos \theta$.
 $\sin 3\theta = 3 \sin \theta \cos^2 \theta - \sin^3 \theta$,
 $= 3 \sin \theta (1 - \sin^2 \theta) - \sin^3 \theta$,
 $= 3 \sin \theta - 3 \sin^3 \theta - \sin^3 \theta$,
 $= 3 \sin \theta - 4 \sin^3 \theta$.

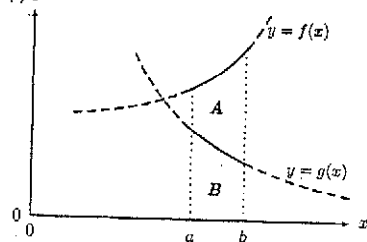
2 (ii) $\cot 3\theta = \frac{\cos 3\theta}{\sin 3\theta}$,
 $= \frac{4 \cos^3 \theta - 3 \cos \theta}{3 \sin \theta - 4 \sin^3 \theta}$,
 $= \frac{4 \cot^3 \theta - 3 \cot \theta \cdot \sec^2 \theta}{3 \sec^2 \theta - 4}$,
 $= \frac{4 \cot^3 \theta - 3 \cot \theta (1 + \cot^2 \theta)}{3(1 + \cot^2 \theta) - 4}$,
 $= \frac{4 \cot^3 \theta - 3 \cot \theta - 3 \cot^3 \theta}{3 + 3 \cot^2 \theta - 4}$,
 $= \frac{\cot^3 \theta - 3 \cot \theta}{3 \cot^2 \theta - 1}$,
 $= \frac{t^3 - 3t}{3t^2 - 1}$, using $t = \cot \theta$.

2 (iii) Now $\cot 3\theta = 1$, $0 \leq \theta \leq 2\pi$
 $3\theta = \frac{\pi}{4}, \frac{5\pi}{4}, \frac{9\pi}{4}$, $0 \leq 3\theta \leq 6\pi \left(\frac{24\pi}{4}\right)$
 $\frac{13\pi}{4}, \frac{17\pi}{4}, \frac{21\pi}{4}$
 $\therefore \theta = \frac{\pi}{12}, \frac{5\pi}{12}, \frac{9\pi}{12}, \frac{13\pi}{12}, \frac{17\pi}{12}, \frac{21\pi}{12}$.

2 (iv) $\frac{t^4 - 3t}{3t^2 - 1} = 1$.
 $\therefore t^4 - 3t^2 - 3t + 1 = 0$.
 As $\cot \theta = \cot(\pi + \theta)$,
 $\theta = \frac{\pi}{12}, \frac{5\pi}{12}, \frac{9\pi}{12}$ are the only distinct values from (iii) above.
 So $t = \cot \theta = \cot \frac{\pi}{12}, \cot \frac{5\pi}{12}, \cot \frac{9\pi}{12}$ are the roots.
 Product of the roots, $-1 = \cot \frac{\pi}{12} \cdot \cot \frac{5\pi}{12} \cdot \cot \frac{9\pi}{12}$.

[1] (v) $\frac{1}{x^3} - \frac{3}{x^2} - \frac{3}{x} + 1 = 0,$
 $x^3 - 3x^2 - 3x + 1 = 0.$

[2] (b) (i) y



$\int_a^b f(x) dx$ is shown by $A + B$ and
 $\int_a^b g(x) dx$ is shown by B .
 It is clear that $A + B > B$,
 i.e., $\int_a^b f(x) dx > \int_a^b g(x) dx$.

[1]

(ii) $y = \tan x,$
 $y' = \sec^2 x > 1 \forall x.$

$\therefore \tan x$ is an increasing function, $\frac{\pi}{4} \leq x \leq \frac{\pi}{3}$ (discontinuities at $\pm \frac{\pi}{2}$ are outside the range).

[3]

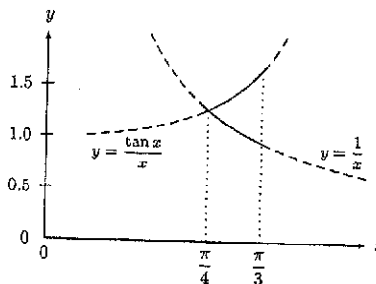
(iii) When $x = \frac{\pi}{4}$, $\tan x = 1$,
 and for $\frac{\pi}{4} < x \leq \frac{\pi}{3}$, $\tan x > 1$ as $\tan x$ is an increasing function.

$\therefore \frac{\tan x}{x} > \frac{1}{x}$ as $x > 0$.

$\therefore$ by part (i): $\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\tan x}{x} dx > \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{1}{x} dx$.

$\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{dx}{x} = (\ln x)_{\frac{\pi}{4}}^{\frac{\pi}{3}},$
 $= \ln \left(\frac{\pi}{3} \cdot \frac{4}{\pi} \right),$
 $= \ln \frac{4}{3}.$

i.e., $\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\tan x}{x} dx > \ln \frac{4}{3}$. [See sketch.]



QUESTION 7

$$\begin{aligned}
 (a) (i) \quad I_n &= \int_1^e x (\ln x)^n dx \\
 &= \int_1^e \frac{d}{dx} \left(\frac{1}{2} x^2 \right) (\ln x)^n dx \\
 &= \left[\frac{1}{2} x^2 (\ln x)^n \right]_1^e - \int_1^e \frac{1}{2} x^2 \cdot n (\ln x)^{n-1} \cdot \frac{1}{x} dx \\
 &= \frac{1}{2} e^2 (\ln e)^n - \frac{1}{2} \cdot 1^2 (\ln 1)^n - \frac{n}{2} \int_1^e x (\ln x)^{n-1} dx \\
 &= \frac{1}{2} e^2 - \frac{1}{2} \cdot 0 - \frac{n}{2} I_{n-1}
 \end{aligned}$$

$$\therefore \boxed{I_n = \frac{1}{2} e^2 - \frac{n}{2} I_{n-1}}$$

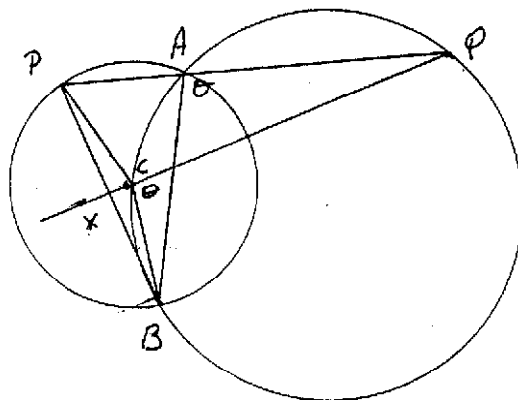
$$\begin{aligned}
 (ii) \quad I_3 &= \frac{e^2}{2} - \frac{3}{2} I_2 \\
 &= \frac{e^2}{2} - \frac{3}{2} \left[\frac{e^2}{2} - I_1 \right] \\
 &= \frac{e^2}{2} - \frac{3e^2}{4} + \frac{3}{2} \left[\frac{e^2}{2} - \frac{1}{2} I_0 \right] \\
 &= \frac{e^2}{2} - \frac{3e^2}{4} + \frac{3e^2}{4} - \frac{3}{4} I_0 \quad ; \quad I_0 = \int_1^e x dx = \left[\frac{x^2}{2} \right]_1^e \\
 &= \frac{e^2}{2} - \frac{3e^2}{4} + \frac{3e^2}{4} - \frac{3}{4} \left(\frac{e^2}{2} - \frac{1}{2} \right) \\
 &= \frac{e^2}{2} - \frac{3e^2}{8} + \frac{3}{8}
 \end{aligned}$$

$$\therefore \boxed{I_3 = \frac{e^2}{8} + \frac{3}{8}}$$

$$\begin{aligned}
 (b) \quad V &= \lim_{\delta x \rightarrow 0} \sum_{x=0}^{\sqrt{6}} 2\pi x y \delta x \\
 &= \int_0^{\sqrt{6}} 2\pi x y dx \\
 &= 2\pi \int_0^{\sqrt{6}} x^3 (6-x^2) dx \\
 &= 2\pi \int_0^{\sqrt{6}} (6x^3 - x^5) dx \\
 &= 2\pi \left[\frac{3x^4}{2} - \frac{x^6}{6} \right]_0^{\sqrt{6}} \\
 &= 2\pi [54 - 36] \\
 &= \boxed{36\pi \text{ m}^3}
 \end{aligned}$$

Q7 (contd)

(c)



(ii) now $\angle BCP = \angle BAP = \theta$ (angles in the same segment standing on the same arc, are equal)

$\therefore \angle BCX = 180^\circ - \theta$ (angles are supplementary)

(iii) now $\angle PAB = 180^\circ - \theta$ (supplementary angles)

$\angle PCB = 360^\circ - 2\theta$ (angle at the centre is double the angle at the circumference standing on same arc)

$\therefore \angle PCX = \angle CBX$ ($180^\circ - \theta + 180^\circ - \theta = 360^\circ - 2\theta$)

CX is common

PC = BC (equal radii)

$\therefore \triangle PCX \cong \triangle BCX$ (SAS)

$\therefore \angle PCX = \angle CBX$ (corresponding angles of congruent Δ 's)

now $\angle PCX + \angle CBX = 180^\circ$ (supplementary angles)

$\therefore \underline{\angle PCX = 90^\circ}$

Question 2.

(a) The roots of $x^3 + ax^2 + bx + c = 0$ are α, β, γ (NB must have 3 roots)

need to establish equation with roots $\alpha\beta, \alpha\gamma + \beta\gamma$.

now $\alpha\beta\gamma = -c$.

$\alpha\beta = -\frac{c}{\gamma} \quad \therefore \text{Let } X = -\frac{c}{\gamma} \text{ i.e. } x = -\frac{c}{\gamma} \text{ in (a)}$

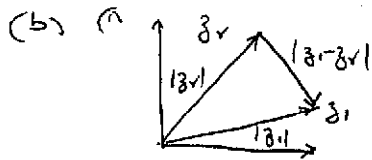
$\left(-\frac{c}{\gamma}\right)^3 + a\left(-\frac{c}{\gamma}\right)^2 + b\left(-\frac{c}{\gamma}\right) + c = 0$

$\frac{-c^3}{\gamma^3} + \frac{ac^2}{\gamma^2} - \frac{bc}{\gamma} + c = 0$

$-c^3 + ac^2\gamma - bc\gamma^2 + c\gamma^3 = 0$

$\Rightarrow x^3 - bx^2 + acx - c^2 = 0$

OR $x^3 - bx^2 + acx - c^2 = 0$



From triangle inequality, "the sum of the lengths of any two sides exceeds the third side."

i.e. $|z - z_v| + |z_v| \geq |z|$

$\therefore |z - z_v| \geq |z| - |z_v|$

(ii) Since $|z - \frac{4}{z}| = 2$ and $|z - \frac{4}{z}| \geq |z| - |\frac{4}{z}|$

then $|z| - \frac{4}{|z|} \leq 2$

$|z|^2 - 4 \leq 2|z|$

$|z|^2 - 2|z| - 4 \leq 0$

$|z|^2 - 2|z| + 1 \leq 5$

$(|z| - 1)^2 \leq 5$

$\therefore -\sqrt{5} \leq |z| - 1 \leq \sqrt{5}$

$-\sqrt{5} + 1 \leq |z| \leq \sqrt{5} + 1$

$\therefore |z| \leq \sqrt{5} + 1 \Rightarrow \text{max. value of } |z| = \sqrt{5} + 1$

8, (c) (i) If $P(x) = 0$ has a root of multiplicity n , say α .

$$\text{then } P(x) = (x - \alpha)^n \cdot Q(x)$$

$$P'(x) = n(x - \alpha)^{n-1} Q(x) + (x - \alpha)^n Q'(x)$$

$$= (x - \alpha)^{n-1} [nQ(x) + (x - \alpha)Q'(x)]$$

now since $Q(x)$ is a polynomial

$nQ(x) + (x - \alpha)Q'(x)$ is a polynomial say $T(x)$.

$$\therefore P'(x) = (x - \alpha)^{n-1} T(x)$$

which has a root α of multiplicity $n-1$.

(ii) Given $x^3 + 3px^2 + 3qx + r = 0$ has a multiple root. (say α)

$$\therefore \alpha^3 + 3p\alpha^2 + 3q\alpha + r = 0 \quad \text{--- (1)}$$

$$+ 3\alpha^2 + 2p\alpha + q = 0 \quad \text{--- (2) from (1)}$$

$$\therefore \alpha^2 + 2p\alpha + q = 0 \quad \text{--- (2a)}$$

$$\text{hence } \alpha^3 + 2p\alpha^2 + q\alpha = 0 \quad \text{--- (3)}$$

From (1) & (3)

$$p\alpha^2 + 2q\alpha + r = 0 \quad \text{--- (4)}$$

$$(2a) \times p$$

$$p\alpha^2 + 2p^2\alpha + pq = 0 \quad \text{--- (2b)}$$

$$(2b) - (4)$$

$$2(p^2 - q)\alpha + pq - r = 0$$

$$\therefore 2(p^2 - q)\alpha = r - pq$$

$$\boxed{\alpha = \frac{r - pq}{2(p^2 - q)}}$$